

Logarithmic Functions & Their Graphs (3.2)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $x = 0$

_____ Evaluate $\log_3 81$.

ANSWER: 7

_____ State the domain of $f(x) = \log(x - 2)$ in interval notation.

ANSWER: 3

_____ Evaluate $\log_2 \frac{1}{16}$.

ANSWER: $\frac{1}{2}$

_____ State the equation of the vertical asymptote of $y = \log_b x$.

ANSWER: $\log_2 32 = 5$ # _____ Write $\log_4 64 = 3$ in exponential form.	ANSWER: 4 # _____ Evaluate $\log_5 1$.
ANSWER: $(2, \infty)$ # _____ Evaluate $\log_9 3$.	ANSWER: -4 # _____ Write $2^5 = 32$ in logarithmic form.
ANSWER: 0 # _____ Evaluate $\log_{10} 1000$.	ANSWER: $4^3 = 64$ # _____ Evaluate $\ln e^7$.